

The Type A_n Case of Keller's Theorem 3.1:

Quiver Mutation, Positive Roots, and Polygon Triangulations

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Abstract

Let Q_n be the linearly oriented Dynkin quiver

$$1 \longrightarrow 2 \longrightarrow \cdots \longrightarrow n.$$

This paper proves the type A_n conclusions of Keller's Theorem 3.1 for the linearly oriented Dynkin quiver Q_n . Starting from the coefficient-free seed with cluster $(x_1, \dots, x_n)$, we show directly that every cluster variable is a Laurent polynomial, that there are exactly $n(n+3)/2$ cluster variables, and that the noninitial variables are naturally indexed by the positive roots

$$\alpha_i + \alpha_{i+1} + \cdots + \alpha_j, \quad 1 \leq i \leq j \leq n.$$

Under this indexing, the reduced denominator is the interval monomial $x_i x_{i+1} \cdots x_j$. The proof combines three descriptions of type A : mutation of quivers, the root system of $\mathfrak{sl}_{n+1}$, and triangulations of an $(n+3)$ -gon. A fan triangulation identifies the support of a denominator with the fan diagonals crossed by the corresponding polygon diagonal. Using the Ptolemy relation, we derive an explicit Laurent expression whose numerator is not divisible by any crossed fan variable. Finally, repeated source sweeps rotate the fan and enumerate every noninitial diagonal, which proves that Keller's knitting algorithm produces all cluster variables in this family. The necessity direction of the general finite-type classification is not addressed.

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1 Introduction

Cluster algebras were introduced by Fomin and Zelevinsky at the beginning of the twenty-first century as a combinatorial framework connected with total positivity and canonical bases in Lie theory [2]; the introductions of Keller and Williams provide the background used here [4, Secs. 1–3] and [5, Secs. 1–2]. Their definition is recursive. One begins with a finite collection of algebraically independent variables and a directed graph, called a quiver. A local operation called *mutation* changes both the graph and one variable. Repeating this operation produces overlapping sets of generators called clusters. The resulting cluster algebra is the subalgebra generated by every variable that appears in this mutation process.

Two features make this recursive definition nontrivial. First, mutation produces rational expressions, so it is not evident from the definition that their denominators remain monomials. The Laurent phenomenon asserts that they do. Second, an unrestricted mutation process may produce infinitely many variables and infinitely many seeds. The finite-type classification says that cluster algebras with finitely many cluster variables are governed by finite root systems, or equivalently by Dynkin diagrams [3]. Keller’s Theorem 3.1 packages the Laurent phenomenon, this classification, the denominator–root correspondence, and the knitting algorithm in the coefficient-free setting [4, Sec. 3.7, Thm. 3.1].

This article is intended for a reader who knows basic linear algebra and the elementary algebra of polynomials and rational functions, but who may know nothing about cluster algebras or root systems. Definitions needed for the main argument are included as they arise. A Dynkin diagram of type A_n is simply the undirected path on n vertices,

$$1 - 2 - \cdots - n.$$

Choosing a direction for each edge turns this graph into a quiver; the orientation studied below directs every edge from left to right.

The purpose of this paper is not to reprove the full classification theorem. We fix the concrete family

$$Q_n : \quad 1 \longrightarrow 2 \longrightarrow \cdots \longrightarrow n \tag{1}$$

and prove the conclusions of Keller’s theorem for this family. This restriction permits an elementary

proof because type A_n has a polygon model. The cluster variables are represented by diagonals of an $(n+3)$ -gon, clusters by triangulations, and mutations by flips. At the same time, the positive roots of A_n are indexed by intervals $[i, j]$. The key point is that a diagonal not belonging to a fixed fan triangulation crosses a consecutive interval of fan diagonals. This makes the root and denominator labels visible in the same picture.

The argument is organized as follows. Section 2 defines quiver and seed mutation. Section 3 computes the types A_2 and A_3 , where the general denominator pattern already appears. Section 4 derives the positive roots of A_n from $\mathfrak{sl}_{n+1}$. Section 5 develops the polygon model and proves locally that flips agree with quiver mutation. Section 6 then proves the main theorem. Its central ingredient is a closed formula obtained from the Ptolemy recurrence; the formula proves that the reduced denominator of a diagonal variable is exactly the product of the fan variables it crosses. The final section summarizes the three equivalent interval structures.

2 Quivers, Seeds, and Mutations

2.1 Quiver mutation

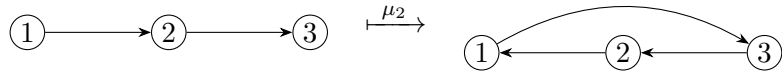
Definition 2.1. A finite *quiver* $Q = (Q_0, Q_1, s, t)$ consists of a finite vertex set Q_0 , a finite arrow set Q_1 , and source and target maps $s, t : Q_1 \rightarrow Q_0$. A *loop* is an arrow whose source equals its target. An *oriented 2-cycle* is a pair of arrows $i \rightarrow j$ and $j \rightarrow i$. Throughout this paper, mutable quivers have no loops and no oriented 2-cycles.

For a vertex k , the mutation $\mu_k(Q)$ is constructed by the standard three-step rule [5, Def. 2.2]:

1. for every oriented path $i \rightarrow k \rightarrow j$, add an arrow $i \rightarrow j$;
2. reverse every arrow incident with k ;
3. cancel a maximal collection of pairwise disjoint oriented 2-cycles.

The last step is unambiguous at the level of the signed number of arrows between two vertices. Two quivers are called *mutation-equivalent* if one can be obtained from the other by a finite sequence of mutations.

Example 2.2. Mutating the path $1 \rightarrow 2 \rightarrow 3$ at its middle vertex gives



The new arrow $1 \rightarrow 3$ comes from the path $1 \rightarrow 2 \rightarrow 3$, while the two arrows incident with 2 are reversed.

The signed adjacency matrix of Q is $B(Q) = (b_{ij})$, where b_{ij} is the number of arrows $i \rightarrow j$ minus the number of arrows $j \rightarrow i$. Quiver mutation is equivalently given by

$$b'_{ij} = \begin{cases} -b_{ij}, & i = k \text{ or } j = k, \\ b_{ij} + [b_{ik}]_+ [b_{kj}]_+ - [-b_{ik}]_+ [-b_{kj}]_+, & i, j \neq k, \end{cases} \quad (2)$$

where $[a]_+ = \max(a, 0)$.

Proposition 2.3. *Quiver mutation is an involution: $\mu_k^2(Q) = Q$.*

Proof. After one mutation, all entries in row or column k change sign. Applying (2) a second time therefore restores those entries. For $i, j \neq k$, the correction term in the first mutation counts the net arrows created by two-paths through k . In the second mutation, $b'_{ik} = -b_{ik}$ and $b'_{kj} = -b_{kj}$, so the correction term is the negative of the first one. Thus $b''_{ij} = b_{ij}$ for all i, j . Equivalently, the arrows incident with k are reversed twice, and the arrows added through k in the first mutation are removed through 2-cycle cancellation in the second. $\square$

A vertex is a *source* if no arrow enters it and a *sink* if no arrow leaves it. Mutation at a source or sink creates no arrows in the first step, because there is no oriented two-path through that vertex. It only reverses the incident arrows. This simple case will be important for Keller's knitting algorithm. Notice that a source can cease to be a source after mutation, while a neighboring vertex can become one. For the linear path, this produces the moving-source pattern

$$1 \rightarrow 2 \rightarrow 3 \rightarrow \cdots \rightarrow n \xrightarrow{\mu_1} 1 \leftarrow 2 \rightarrow 3 \rightarrow \cdots \rightarrow n \xrightarrow{\mu_2} 1 \rightarrow 2 \leftarrow 3 \rightarrow \cdots \rightarrow n.$$

Continuing from left to right makes vertex k a source immediately before mutation at k .

2.2 Seeds and cluster algebras

Let

$$\mathcal{F} = \mathbb{Q}(x_1, \dots, x_n)$$

be the field of rational functions in n algebraically independent variables.

Definition 2.4. A coefficient-free labeled *seed* is a pair $\Sigma = (Q, \mathbf{x})$, where Q is a quiver on vertices $1, \dots, n$ and $\mathbf{x} = (x_1, \dots, x_n)$ is a transcendence basis of $\mathcal{F}$. Concretely, this means that the x_i satisfy no nonzero polynomial relation over $\mathbb{Q}$ and that every element of $\mathcal{F}$ is a rational function in them. The ordered tuple $\mathbf{x}$ is the *cluster* of the seed.

Mutation of a seed at k replaces Q by $\mu_k(Q)$ and replaces only the variable x_k . The new variable x'_k is determined by the exchange relation

$$x_k x'_k = \prod_{i \rightarrow k} x_i + \prod_{k \rightarrow j} x_j, \quad \prod_{\emptyset} = 1. \quad (3)$$

Products are taken over arrows, so parallel arrows contribute powers. The other cluster variables remain unchanged. Seed mutation is also involutive: the quiver returns by Proposition 2.3, and the second exchange relation solves back to x_k .

Example 2.5 (Mutation of a seed). For the initial seed with quiver $1 \rightarrow 2 \rightarrow 3$ and cluster (x_1, x_2, x_3) , mutation at 2 changes the quiver as in Example 2.2 and gives

$$x'_2 = \frac{x_1 + x_3}{x_2}.$$

Indeed, the incoming product is x_1 and the outgoing product is x_3 . The mutated seed is

$$\left(\begin{array}{c} \text{Quiver: } 1 \leftarrow 2 \leftarrow 3 \\ \text{Cluster: } \left(x_1, \frac{x_1 + x_3}{x_2}, x_3 \right) \end{array} \right).$$

Mutating at 2 again uses the reversed incident arrows. The two monomials in the exchange relation are still x_1 and x_3 , so

$$x_2'' = \frac{x_1 + x_3}{x_2'} = x_2.$$

This calculation illustrates simultaneously the quiver and variable parts of the involution.

Definition 2.6. Starting from an initial seed $\Sigma_0 = (Q, \mathbf{x})$, a *cluster variable* is any variable appearing in a seed obtained by a finite mutation sequence. A *cluster* is the set of variables in such a seed. The coefficient-free cluster algebra associated with Q is

$$\mathcal{A}_Q = \mathbb{Q}[\text{all cluster variables}] \subseteq \mathcal{F}.$$

The *exchange graph* has seeds (or unlabeled clusters, when no confusion results) as vertices and mutations as edges.

A *Laurent polynomial* is a finite $\mathbb{Q}$ -linear combination of monomials $x_1^{a_1} \cdots x_n^{a_n}$ in which the exponents a_i may be negative. Equivalently, its denominator can be chosen to be a monomial.

Definition 2.7 (Reduced denominator). A Laurent polynomial $X \in \mathbb{Q}[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$ has a unique expression

$$X = \frac{P(x_1, \dots, x_n)}{x_1^{d_1} \cdots x_n^{d_n}}, \quad d_i \geq 0,$$

where $P \in \mathbb{Q}[x_1, \dots, x_n]$ and, whenever $d_i > 0$, the polynomial P is not divisible by x_i . We call

$$\text{den}(X) = x_1^{d_1} \cdots x_n^{d_n}$$

the *reduced denominator monomial* of X , and

$$\text{dvec}(X) = (d_1, \dots, d_n)$$

its *denominator vector*. The adjective “reduced” means that no factor x_i can be cancelled between the numerator and denominator.

The distinction between subalgebra and subfield is essential. The ambient field contains arbitrary rational functions, whereas $\mathcal{A}_Q$ is generated under addition and multiplication by the distinguished mutation variables. In general, most elements of $\mathcal{F}$ are not in $\mathcal{A}_Q$.

Remark 2.8 (Coefficient convention). Williams’s polygon model uses an extended seed: polygon diagonals are mutable and polygon sides are frozen coefficient variables [5, Sec. 2.2]. Keller’s Section 3 uses the coefficient-free convention [4, Secs. 3.2–3.4]. We first use the polygon model with side variables in order to see the Ptolemy exchange relation, and then apply the specialization sending every boundary-side variable to 1. Because mutation relations are polynomial identities in the coefficient variables, this specialization commutes with every mutation. All denominator statements below refer to the resulting coefficient-free variables.

3 The Examples of Types A_2 and A_3

The smallest cases display the three phenomena that the main theorem will explain: Laurent denominators, finiteness, and positive-root indexing [4, Secs. 2.2, 3.5–3.6].

3.1 Type A_2

Begin with $Q_2 : 1 \rightarrow 2$ and cluster (x_1, x_2) . Mutating alternately at 1 and 2 gives

$$\begin{aligned} x'_1 &= \frac{1+x_2}{x_1}, \\ x'_2 &= \frac{1+x'_1}{x_2} = \frac{1+x_1+x_2}{x_1x_2}, \\ x''_1 &= \frac{1+x'_2}{x'_1} = \frac{1+x_1}{x_2}. \end{aligned}$$

The next mutation gives

$$\frac{1+x''_1}{x'_2} = x_1,$$

and the following one gives x_2 , so the pattern repeats. Hence the five cluster variables are

$$x_1, \quad x_2, \quad \frac{1+x_2}{x_1}, \quad \frac{1+x_1+x_2}{x_1x_2}, \quad \frac{1+x_1}{x_2}. \quad (4)$$

The calculation can be recorded seed by seed as follows. The arrows alternate because every mutation reverses the unique arrow. Modulo simultaneous relabeling of the vertices and cluster variables, the five seeds form a pentagon in the exchange graph.

Table 1: A complete mutation cycle for A_2 .

Step	Quiver before mutation	Direction	New variable
0	$1 \rightarrow 2$	1	$x'_1 = (1+x_2)/x_1$
1	$1 \leftarrow 2$	2	$x'_2 = (1+x_1+x_2)/(x_1x_2)$
2	$1 \rightarrow 2$	1	$x''_1 = (1+x_1)/x_2$
3	$1 \leftarrow 2$	2	x_1
4	$1 \rightarrow 2$	1	x_2

The reduced denominator vectors of the three noninitial variables are

$$(1, 0), \quad (1, 1), \quad (0, 1),$$

which match the positive roots α_1 , $\alpha_1 + \alpha_2$, and α_2 of A_2 .

3.2 Type A_3

Now start from $1 \rightarrow 2 \rightarrow 3$. A source mutation at 1, followed by source mutations at 2 and 3, gives

$$u_{1,1} = \frac{1+x_2}{x_1}, \quad (5)$$

$$u_{1,2} = \frac{1+u_{1,1}x_3}{x_2} = \frac{x_1+x_3+x_2x_3}{x_1x_2}, \quad (6)$$

$$u_{1,3} = \frac{1+u_{1,2}}{x_3} = \frac{x_1x_2+x_1+x_3+x_2x_3}{x_1x_2x_3}. \quad (7)$$

A direct mutation at the middle vertex gives

$$u_{2,2} = \frac{x_1 + x_3}{x_2}.$$

After this mutation, mutating at 3 gives

$$u_{2,3} = \frac{x_1 + u_{2,2}}{x_3} = \frac{x_1x_2 + x_1 + x_3}{x_2x_3}.$$

Finally, direct mutation at 3 gives

$$u_{3,3} = \frac{1 + x_2}{x_3}.$$

Together with x_1, x_2, x_3 , these are the nine cluster variables. The formulas are not an arbitrary list. The first three come from the source sequence 1, 2, 3 and have denominator intervals beginning at 1. The next two have intervals beginning at 2, and the last begins at 3. In particular, the six denominator vectors are

$$(1, 0, 0), (1, 1, 0), (1, 1, 1), (0, 1, 0), (0, 1, 1), (0, 0, 1).$$

Their denominator data are summarized in Table 2.

Table 2: The six noninitial variables of type A_3 .

Variable	Reduced denominator	Positive root
$u_{1,1}$	x_1	α_1
$u_{1,2}$	x_1x_2	$\alpha_1 + \alpha_2$
$u_{1,3}$	$x_1x_2x_3$	$\alpha_1 + \alpha_2 + \alpha_3$
$u_{2,2}$	x_2	α_2
$u_{2,3}$	x_2x_3	$\alpha_2 + \alpha_3$
$u_{3,3}$	x_3	α_3

Thus the noninitial denominators are precisely the consecutive interval monomials $x_i \cdots x_j$. There are $3 + 6 = 9$ cluster variables. The examples suggest that for general A_n , positive roots, noninitial variables, and intervals $[i, j]$ should all be the same indexing set.

4 The Root System of Type A_n

This section derives the required root data from $\mathfrak{sl}_{n+1}$. Only the matrix Lie algebra, its Cartan subalgebra, and its root decomposition are needed; no general Lie group theory will be used. The presentation follows Etingof's discussion of Cartan subalgebras, root decompositions, classical root systems, and positive roots [1, Lects. 17–19].

4.1 The roots of $\mathfrak{sl}_{n+1}$

A complex Lie algebra is a complex vector space equipped with a bilinear bracket satisfying skew-symmetry and the Jacobi identity. For matrices the bracket is the commutator

$$[X, Y] = XY - YX.$$

The Lie algebra

$$\mathfrak{sl}_{n+1}(\mathbb{C}) = \{X \in M_{n+1}(\mathbb{C}) : \text{tr}(X) = 0\}$$

is closed under this bracket. It acts on itself by the adjoint representation, $\text{ad}(X)(Y) = [X, Y]$.

Let $\mathfrak{h} \subset \mathfrak{sl}_{n+1}$ be the diagonal trace-zero subalgebra. Write

$$h = \text{diag}(h_1, \dots, h_{n+1}), \quad \sum_{r=1}^{n+1} h_r = 0,$$

and define $\varepsilon_i \in \mathfrak{h}^*$ by $\varepsilon_i(h) = h_i$. If E_{ij} is the matrix unit with a 1 in position (i, j) , then

$$[h, E_{ij}] = hE_{ij} - E_{ij}h = (h_i - h_j)E_{ij} = (\varepsilon_i - \varepsilon_j)(h)E_{ij}. \quad (8)$$

Therefore $\mathbb{C}E_{ij}$ is a simultaneous eigenspace for $\text{ad}(\mathfrak{h})$ with weight $\varepsilon_i - \varepsilon_j$. The root system is

$$\Phi(A_n) = \{\varepsilon_i - \varepsilon_j : 1 \leq i \neq j \leq n+1\}. \quad (9)$$

It lies in the n -dimensional hyperplane of $\mathbb{R}^{n+1}$ whose coordinate sum is zero.

For context, a reduced crystallographic root system in a Euclidean space is a finite spanning set Φ such that $0 \notin \Phi$, $\Phi \cap \mathbb{R}\alpha = \{\alpha, -\alpha\}$ for every $\alpha \in \Phi$, reflection in the hyperplane perpendicular to any root preserves Φ , and the Cartan integers $2(\alpha, \beta)/(\alpha, \alpha)$ are integral. A choice of a generic linear functional separates Φ into positive and negative roots. A positive root is *simple* if it is not a sum of two positive roots. General root-system theory shows that the simple roots form a basis and every positive root is a nonnegative integral combination of them [1, Lect. 19, Sec. 19.3].

Choose the positive system in which $\varepsilon_i - \varepsilon_j$ is positive exactly when $i < j$. The simple roots are

$$\alpha_i = \varepsilon_i - \varepsilon_{i+1}, \quad 1 \leq i \leq n. \quad (10)$$

To see this directly, if $j \geq i+2$, then

$$\varepsilon_i - \varepsilon_j = (\varepsilon_i - \varepsilon_{i+1}) + (\varepsilon_{i+1} - \varepsilon_j)$$

is a sum of two positive roots, so it is not simple. On the other hand, $\varepsilon_i - \varepsilon_{i+1}$ cannot be decomposed into two positive roots: an intermediate index strictly between i and $i+1$ does not exist. Thus the roots in (10) are exactly the simple roots. Their inner products satisfy $(\alpha_i, \alpha_{i+1}) = -1$, while nonadjacent simple roots are orthogonal, so their Dynkin diagram is the path A_n .

The complete root-space decomposition is

$$\mathfrak{sl}_{n+1} = \mathfrak{h} \oplus \bigoplus_{i \neq j} \mathbb{C}E_{ij}.$$

The zero-weight space is $\mathfrak{h}$, and every nonzero weight is one of the roots in (9). This matrix decomposition is the Lie-theoretic reason that type A_n appears with rank n even though the matrices have size $n + 1$.

Proposition 4.1. *The positive roots of type A_n are exactly*

$$\Phi^+(A_n) = \{\alpha_i + \alpha_{i+1} + \cdots + \alpha_j : 1 \leq i \leq j \leq n\}.$$

Consequently, $|\Phi^+(A_n)| = n(n + 1)/2$.

Proof. For $1 \leq i < k \leq n + 1$, the sum telescopes:

$$\alpha_i + \alpha_{i+1} + \cdots + \alpha_{k-1} = (\varepsilon_i - \varepsilon_{i+1}) + \cdots + (\varepsilon_{k-1} - \varepsilon_k) = \varepsilon_i - \varepsilon_k.$$

Every positive root in (9) has the form $\varepsilon_i - \varepsilon_k$ with $i < k$, so every positive root is such a consecutive sum. Conversely, every consecutive sum is $\varepsilon_i - \varepsilon_{j+1}$ and is positive. The roots are therefore indexed by intervals $[i, j] \subseteq [1, n]$. There are $n - i + 1$ intervals beginning at i , or equivalently

$$n + (n - 1) + \cdots + 1 = \frac{n(n + 1)}{2}$$

intervals in total. □

The interval description in Proposition 4.1 is exactly the pattern observed in the denominator tables for A_2 and A_3 .

5 Polygon Triangulations and Type A

5.1 Triangulations and their quivers

Let P_d be a convex polygon with d labeled vertices. A diagonal joins two nonadjacent vertices. A triangulation is a maximal collection of pairwise noncrossing diagonals.

Lemma 5.1. *Every triangulation of a d -gon contains $d - 3$ internal diagonals and $d - 2$ triangles.*

Proof. Let r be the number of internal diagonals and t the number of triangles. The dual graph of a polygon triangulation is a tree: its vertices are the triangles, and two are adjacent when they share an internal diagonal. Hence $r = t - 1$. Counting triangle-edge incidences gives $3t = d + 2r$, since each boundary side occurs once and each internal diagonal twice. Substituting $r = t - 1$ yields $t = d - 2$ and $r = d - 3$. □

Thus rank n corresponds to a polygon with $d = n + 3$ vertices.

Following Williams, associate an extended quiver $Q(T)$ to a triangulation T [5, Def. 2.12]. Put a mutable vertex at the midpoint of every internal diagonal and a frozen vertex at the midpoint of every boundary side. Inside each triangle, connect its three midpoint vertices in a clockwise oriented 3-cycle. Arrows between two frozen vertices may be suppressed, since they do not affect mutation. The principal part consists only of the mutable vertices.

For the fan triangulation based at vertex 0 of an $(n + 3)$ -gon with counterclockwise vertices $0, 1, \dots, n + 2$, set

$$\tau_i = (0, i + 1), \quad 1 \leq i \leq n. \quad (11)$$

With the clockwise-arrow convention, the mutable arrows are

$$\tau_1 \longrightarrow \tau_2 \longrightarrow \cdots \longrightarrow \tau_n.$$

Thus the principal quiver is precisely Q_n in (1).

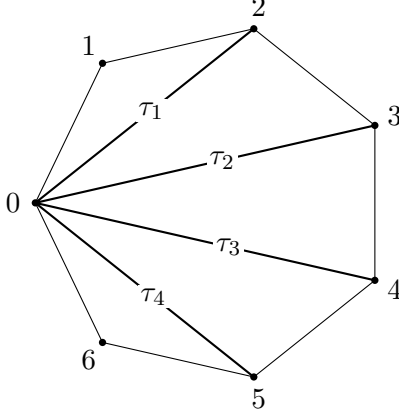


Figure 1: The fan triangulation for $n = 4$. Its mutable quiver is $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$.

5.2 Flips are mutations

Every internal diagonal k belongs to exactly two triangles, whose union is a quadrilateral. Replacing k by the other diagonal k' is called a flip.

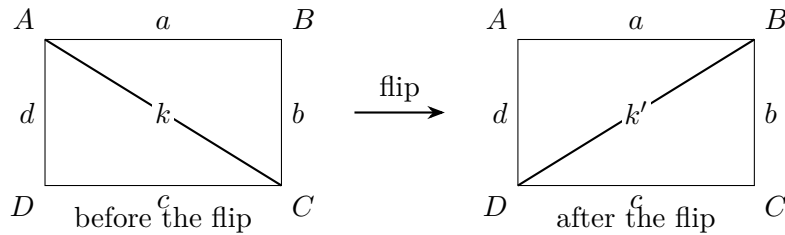


Figure 2: A flip replaces one diagonal of a quadrilateral by the other. Using the same symbols for the sides and diagonals and for their associated variables, the exchange relation is $kk' = ac + bd$, with ac and bd corresponding to the two pairs of opposite sides.

Proposition 5.2. *If T' is obtained from T by flipping an internal diagonal k , then*

$$Q(T') = \mu_k(Q(T))$$

after identifying the new diagonal k' with the mutated vertex k .

Proof. Only the two triangles adjacent to k can change the quiver. Use the notation of Figure 2: label the quadrilateral vertices A, B, C, D counterclockwise, let $k = AC$, and write

$$a = AB, \quad b = BC, \quad c = CD, \quad d = DA.$$

Before the flip, the clockwise 3-cycles in triangles ABC and ACD give

$$k \rightarrow b \rightarrow a \rightarrow k, \quad k \rightarrow d \rightarrow c \rightarrow k.$$

Thus the two-paths through k are

$$a \rightarrow k \rightarrow b, \quad a \rightarrow k \rightarrow d, \quad c \rightarrow k \rightarrow b, \quad c \rightarrow k \rightarrow d.$$

Mutation adds $a \rightarrow b$, $a \rightarrow d$, $c \rightarrow b$, and $c \rightarrow d$; it reverses the four arrows incident with k . The added arrows $a \rightarrow b$ and $c \rightarrow d$ cancel the existing arrows $b \rightarrow a$ and $d \rightarrow c$. The surviving local arrows are

$$k' \rightarrow a, \quad k' \rightarrow c, \quad b \rightarrow k', \quad d \rightarrow k', \quad a \rightarrow d, \quad c \rightarrow b.$$

After the flip, the new triangles are BCD and BDA . Their clockwise cycles are

$$k' \rightarrow c \rightarrow b \rightarrow k', \quad d \rightarrow k' \rightarrow a \rightarrow d,$$

which give exactly the same arrows. The argument remains valid when some of a, b, c, d are boundary sides: those vertices are frozen, and any arrows between two frozen vertices are simply ignored. $\square$

Lemma 5.3. *The flip graph of a convex polygon is connected.*

Proof. Fix a vertex v of the polygon. We show that every triangulation can be transformed into the fan based at v . Count the internal diagonals of a triangulation that are incident with v . If there are $d - 3$ of them, the triangulation is already the fan based at v .

Otherwise, list in cyclic order the vertices joined to v by a boundary side or an internal diagonal. Two consecutive vertices a and b in this list are not adjacent along the boundary for at least one such pair. Since no diagonal from v ends strictly between a and b , the triangle of the triangulation incident with v in that sector is (v, a, b) , and (a, b) is an internal diagonal. Let c be the third vertex of the other triangle incident with (a, b) . Flipping (a, b) replaces it by (v, c) . The new diagonal is incident with v and was not present before, so the number of internal diagonals incident with v increases by one.

Repeating this operation eventually produces the fan based at v . Hence every triangulation is connected by flips to the same fan, and the flip graph is connected. $\square$

Consequently, all triangulations of a fixed d -gon yield mutation-equivalent seeds and isomorphic cluster algebras [5, Prop. 2.17].

5.3 Plücker coordinates and diagonals as cluster variables

The Ptolemy relation used in the polygon model is not an additional assumption: it is a two-dimensional determinant identity. Let

$$v_i = \begin{pmatrix} a_i \\ b_i \end{pmatrix} \in \mathbb{C}^2, \quad p_{ij} = \det(v_i, v_j) = a_i b_j - a_j b_i \quad (i < j).$$

The quantities p_{ij} are the Plücker coordinates for the affine cone over the Grassmannian $\text{Gr}_{2,d}$. This terminology is included only for context: no knowledge of Grassmannians is required, and the argument below uses only the determinant identity in Lemma 5.4.

Lemma 5.4 (Three-term Plücker relation). *For cyclically ordered indices $i < j < k < \ell$,*

$$p_{ik}p_{j\ell} = p_{ij}p_{k\ell} + p_{i\ell}p_{jk}. \quad (12)$$

Proof. Expand the right-hand side using $p_{rs} = a_r b_s - a_s b_r$. After two pairs of equal terms cancel, the result factors as

$$a_i a_j b_k b_\ell - a_i a_\ell b_j b_k - a_j a_k b_i b_\ell + a_k a_\ell b_i b_j = (a_i b_k - a_k b_i)(a_j b_\ell - a_\ell b_j) = p_{ik}p_{j\ell}.$$

Thus (12) holds identically. $\square$

When ik and $j\ell$ are the two diagonals of a quadrilateral, (12) has the form

$$(\text{old diagonal})(\text{new diagonal}) = (\text{one pair of opposite sides}) + (\text{the other pair}).$$

It is therefore precisely the seed exchange relation for a flip.

Theorem 5.5 (Polygon model for type A). *Let T be a triangulation of a d -gon and let $Q(T)$ be its extended quiver. The cluster algebra determined by this seed has the following interpretation:*

- (a) *boundary sides correspond to frozen variables and internal diagonals correspond to mutable cluster variables;*
- (b) *triangulations correspond to extended clusters;*
- (c) *a flip of a diagonal corresponds to mutation at the associated mutable vertex;*
- (d) *under these identifications, the exchange relation is the three-term Plücker relation (12).*

Consequently the isomorphism class depends only on d , and the resulting cluster algebra has type A_{d-3} .

Proof. Williams constructs $Q(T)$ in Definition 2.12 and the associated seed in Definition 2.13 [5, Defs. 2.12–2.13]. The local compatibility of flips and quiver mutation is proved directly in Proposition 5.2; the variable part is (12), proved in Lemma 5.4. Since the flip graph is connected by Lemma 5.3, every triangulation is reached from every other one, so the cluster algebra is independent of the initial triangulation. Finally, every diagonal belongs to some triangulation: the diagonal divides the polygon into two smaller polygons, and triangulating both pieces gives a triangulation containing it. Hence all diagonals occur as mutable variables, all triangulations occur as clusters, and

flips give all adjacent seeds. This is the polygon model summarized in Williams's Proposition 2.17 and Exercise 2.18 [5, Prop. 2.17 and Ex. 2.18]. $\square$

We now pass from Williams's extended seed to Keller's coefficient-free convention. Fix a mutation word. Start with the same mutable fan variables in both models, but put an independent frozen variable on each boundary side in the extended model and put the value 1 on every boundary side in the coefficient-free model. At every mutation step the two exchange relations have the same two products; evaluating the frozen side variables at 1 changes the extended relation into the coefficient-free relation. Induction on the length of the mutation word therefore shows that specialization commutes with the entire mutation sequence.

Thus every coefficient-free cluster variable is obtained by specializing a polygon diagonal variable. Conversely, every diagonal can be reached by flips by Theorem 5.5, and its specialization is a coefficient-free cluster variable. A d -gon has

$$\binom{d}{2} - d = \frac{d(d-3)}{2}$$

diagonals. With $d = n + 3$, this becomes

$$\frac{n(n+3)}{2}. \tag{13}$$

At this stage, specialization could in principle identify two different diagonal variables. The reduced-denominator formula in the next section will rule this out.

Remark 5.6 (Number of clusters). The polygon model also identifies clusters with triangulations. Thus the number of clusters of type A_n is the classical Catalan number

$$C_{n+1} = \frac{1}{n+2} \binom{2n+2}{n+1}.$$

This supplementary enumeration is not used in the proof of Theorem 6.1; the theorem concerns cluster variables, which are counted instead by the diagonals in (13).

6 Proof of Keller's Theorem 3.1 in Type A_n

We can now state the precise result proved in this paper.

Theorem 6.1 (Type A_n case of Keller's Theorem 3.1). *Let Q_n be the linearly oriented quiver (1), with initial cluster $(x_1, \dots, x_n)$. Then:*

- (i) *every cluster variable is a Laurent polynomial in $x_1, \dots, x_n$;*
- (ii) *the cluster algebra has exactly $n(n+3)/2$ distinct cluster variables;*
- (iii) *the noninitial cluster variables are naturally in bijection with the positive roots*

$$\alpha_i + \alpha_{i+1} + \dots + \alpha_j, \quad 1 \leq i \leq j \leq n;$$

- (iv) *the variable corresponding to $\alpha_i + \dots + \alpha_j$ has reduced denominator monomial $x_i x_{i+1} \dots x_j$;*

(v) *Keller's knitting algorithm produces every cluster variable.*

This is the sufficiency and explicit-description part of Keller's theorem for one Dynkin family. It does not prove that an arbitrary cluster-finite quiver must be mutation-equivalent to a Dynkin orientation.

6.1 Crossing intervals and positive roots

Fix the fan triangulation (11). Every diagonal not in the fan has both endpoints among $1, 2, \dots, n+2$. Write it as (a, b) with

$$1 \leq a < b \leq n+2, \quad b-a \geq 2.$$

Lemma 6.2. *The diagonal (a, b) crosses precisely the fan diagonals*

$$\tau_a, \tau_{a+1}, \dots, \tau_{b-2}.$$

Proof. In a convex polygon, two diagonals cross exactly when their endpoints alternate around the boundary. The fan diagonal $\tau_r = (0, r+1)$ crosses (a, b) exactly when

$$a < r+1 < b,$$

which is equivalent to $a \leq r \leq b-2$. □

Define the crossing vector $\mathbf{c}(a, b) = (c_1, \dots, c_n)$ by $c_r = 1$ if (a, b) crosses τ_r and $c_r = 0$ otherwise. By Lemma 6.2, it is the indicator vector of the interval $[a, b-2]$.

Proposition 6.3. *The map*

$$(a, b) \mapsto \alpha_a + \alpha_{a+1} + \dots + \alpha_{b-2} \tag{14}$$

is a bijection from the nonfan diagonals to $\Phi^+(A_n)$.

Proof. By Proposition 4.1, every positive root is uniquely indexed by an interval $[i, j]$ with $1 \leq i \leq j \leq n$. The inverse of (14) sends $\alpha_i + \dots + \alpha_j$ to the diagonal $(i, j+2)$. Its endpoints satisfy $1 \leq i < j+2 \leq n+2$, and their difference is at least two. Hence it is a nonfan diagonal. The two constructions are inverse. □

6.2 An explicit denominator formula

The main technical point is to prove that the crossing interval is not merely a label but the exact reduced denominator vector. We use the Ptolemy recurrence to obtain a closed Laurent expression.

Let $X_{a,b}$ denote the coefficient-free cluster variable represented by the diagonal (a, b) . For convenience define

$$y_r = p_{0r}|_{\text{boundary}=1} = \begin{cases} 1, & r = 1 \text{ or } r = n+2, \\ x_{r-1}, & 2 \leq r \leq n+1. \end{cases} \tag{15}$$

Thus the fan variable x_i is y_{i+1} . Boundary edges satisfy $X_{a,a+1} = 1$.

Apply (12) to the quadrilateral with vertices $0, a, b-1, b$. Since $(b-1, b)$ is a boundary edge, specialization gives

$$y_{b-1}X_{a,b} = y_a + y_bX_{a,b-1}, \quad b \geq a+2. \tag{16}$$

For $b \geq a + 1$, set

$$D_{a,b} = \prod_{r=a+1}^{b-1} y_r, \quad (17)$$

$$N_{a,b} = \sum_{k=a}^{b-1} \left(\prod_{r=a}^{k-1} y_r \right) \left(\prod_{r=k+2}^b y_r \right), \quad (18)$$

where every empty product is 1. For example,

$$N_{a,a+2} = y_a + y_{a+2},$$

and

$$N_{a,a+3} = y_a y_{a+1} + y_a y_{a+3} + y_{a+2} y_{a+3}.$$

Proposition 6.4 (Denominator formula). *For every nonfan diagonal (a, b) ,*

$$X_{a,b} = \frac{N_{a,b}}{D_{a,b}}. \quad (19)$$

This fraction is reduced with respect to the initial variables $x_1, \dots, x_n$. Consequently,

$$\text{den}(X_{a,b}) = \prod_{r=a+1}^{b-1} y_r = x_a x_{a+1} \cdots x_{b-2}. \quad (20)$$

Proof. We first prove (19) by induction on $b - a$. When $b = a + 1$, both sides equal 1: $D_{a,a+1} = 1$ and the single summand in $N_{a,a+1}$ is 1. Suppose the formula holds for $X_{a,b-1}$. Equation (16) gives

$$X_{a,b} = \frac{y_a + y_b N_{a,b-1} / D_{a,b-1}}{y_{b-1}} = \frac{y_a D_{a,b-1} + y_b N_{a,b-1}}{y_{b-1} D_{a,b-1}}.$$

The denominator is $D_{a,b}$ by (17). In the numerator, the term

$$y_a D_{a,b-1} = \prod_{r=a}^{b-2} y_r$$

is exactly the $k = b - 1$ summand of $N_{a,b}$. Multiplying $N_{a,b-1}$ by y_b gives the summands indexed by $k = a, \dots, b - 2$. Hence the numerator is $N_{a,b}$, completing the induction.

It remains to prove that no denominator factor cancels. Let s satisfy $a + 1 \leq s \leq b - 1$. In the sum (18), choose the term indexed by $k = s - 1$. That term omits the two factors y_{s-1} and y_s , so in particular it is not divisible by y_s . Every coefficient in $N_{a,b}$ is positive. Specializing the boundary values y_1 and y_{n+2} to 1 can merge monomials and add positive coefficients, but it cannot create cancellation. Therefore $N_{a,b}$ is not divisible by any y_s appearing in $D_{a,b}$. Since the interior y_s are distinct initial variables, the fraction is reduced. Substituting $y_r = x_{r-1}$ gives (20). $\square$

Example 6.5. For a diagonal crossing three consecutive fan arcs, say $(a, a + 4)$, the formula gives

$$X_{a,a+4} = \frac{y_a y_{a+1} y_{a+2} + y_a y_{a+1} y_{a+4} + y_a y_{a+3} y_{a+4} + y_{a+2} y_{a+3} y_{a+4}}{y_{a+1} y_{a+2} y_{a+3}}.$$

Each denominator variable is absent from at least one numerator monomial, so none can cancel. For A_3 , taking $(a, b) = (1, 5)$ and $y_1 = y_5 = 1$ recovers (7).

6.3 The Laurent property, finiteness, and the root correspondence

By Theorem 5.5, every mutation sequence in type A_n is represented by flips, and the specialization argument above commutes with the exchange relations. Thus every coefficient-free cluster variable is the specialization of a diagonal variable, and every polygon diagonal produces a coefficient-free cluster variable.

Corollary 6.6. *Statements (i)–(iv) of Theorem 6.1 hold.*

Proof. The initial variables are Laurent monomials. Every noninitial variable is $X_{a,b}$ for a nonfan diagonal, and Proposition 6.4 gives an explicit Laurent expression. This proves (i).

The nonfan diagonals are in bijection with positive roots by Proposition 6.3. Under this bijection, Proposition 6.4 sends

$$\alpha_a + \cdots + \alpha_{b-2} \longleftrightarrow X_{a,b} \text{ with denominator } x_a \cdots x_{b-2}.$$

This proves (iii) and (iv). Different nonfan diagonals have different denominator intervals, so their specialized variables remain distinct. They are also distinct from the initial variables because their reduced denominators are nontrivial. There are $n(n+1)/2$ positive roots and n initial variables, hence

$$n + \frac{n(n+1)}{2} = \frac{n(n+3)}{2}$$

distinct cluster variables. This proves (ii). $\square$

This argument proves the Laurent property directly in type A_n , rather than invoking the general Laurent phenomenon. It also explains why the denominator coefficients are only 0 and 1: a polygon diagonal crosses any fixed fan diagonal at most once.

6.4 The repetition quiver and the knitting algorithm

We first give a self-contained description of Keller's knitting algorithm for the linearly oriented quiver Q_n . The repetition quiver $\mathbb{Z}Q_n$ has vertices

$$(m, i), \quad m \in \mathbb{Z}, \quad 1 \leq i \leq n,$$

and, for each arrow $i \rightarrow i+1$ of Q_n , arrows

$$(m, i) \longrightarrow (m, i+1), \quad (m, i+1) \longrightarrow (m+1, i).$$

Thus the immediate predecessors of (m, i) are $(m, i-1)$ and $(m-1, i+1)$ whenever these vertices exist. Put

$$z_{0,i} = x_i, \quad z_{m,0} = z_{m,n+1} = 1 \quad (m \geq 0),$$

and compute the remaining variables from left to right by the knitting recurrence

$$z_{m,i} z_{m-1,i} = 1 + z_{m,i-1} z_{m-1,i+1}, \quad m \geq 1, \quad 1 \leq i \leq n. \quad (21)$$

This is Keller's predecessor-product rule specialized to the repetition quiver of the linear path [4, Secs. 2.2 and 3.7].

Lemma 6.7. *For each $m \geq 1$, the variables*

$$z_{m,1}, z_{m,2}, \dots, z_{m,n}$$

are obtained from the cluster $(z_{m-1,1}, \dots, z_{m-1,n})$ by mutating successively at the sources

$$1, 2, \dots, n.$$

Consequently, the knitting recurrence and repeated left-to-right source sweeps produce the same cluster variables.

Proof. At the beginning of a sweep the quiver is $1 \rightarrow 2 \rightarrow \dots \rightarrow n$. A full left-to-right sweep restores this orientation, because each edge is reversed once when its left endpoint is mutated and once when its right endpoint is mutated. After mutations at $1, \dots, i-1$ within the current sweep, the vertex i is a source: the arrow joining it to $i-1$ points from i to $i-1$, while the arrow joining it to $i+1$ still points from i to $i+1$. Immediately before mutation at i , the variable at $i-1$ has already become $z_{m,i-1}$, the variable at $i+1$ is still $z_{m-1,i+1}$, and the variable at i is $z_{m-1,i}$. Since the incoming product at a source is the empty product, the exchange relation is

$$z_{m-1,i} z_{m,i} = 1 + z_{m,i-1} z_{m-1,i+1},$$

with the boundary conventions in (21). This is exactly the knitting recurrence. The assertion follows by induction on i , and then on m . $\square$

We now identify these sweeps in the polygon model. In the mutation word

$$\mu_n \cdots \mu_2 \mu_1, \tag{22}$$

operations are performed from right to left, so μ_1 is applied first. By the orientation argument in the proof of Lemma 6.7, the same source sweep can be repeated indefinitely.

The first sweep flips

$$(0, 2), (0, 3), \dots, (0, n+1)$$

in this order and changes the fan based at 0 into the fan based at 1. At the first step, $(0, 2)$ lies in the quadrilateral with vertices 0, 1, 2, 3 and is replaced by $(1, 3)$. After that flip, $(0, 3)$ lies in a quadrilateral whose opposite diagonal is $(1, 4)$. Continuing inductively, the new diagonal introduced at the k th step is $(1, k+2)$, which crosses $\tau_1, \dots, \tau_k$ and has denominator $x_1 \cdots x_k$.

The geometric effect of one full sweep is shown in Figure 3: the fan center moves from one boundary vertex to the next.

For example, in type A_4 the genuinely new variables appear in the triangular array

sweep 1	[1, 1]	[1, 2]	[1, 3]	[1, 4]
sweep 2	[2, 2]	[2, 3]	[2, 4]	
sweep 3	[3, 3]	[3, 4]		
sweep 4	[4, 4]			

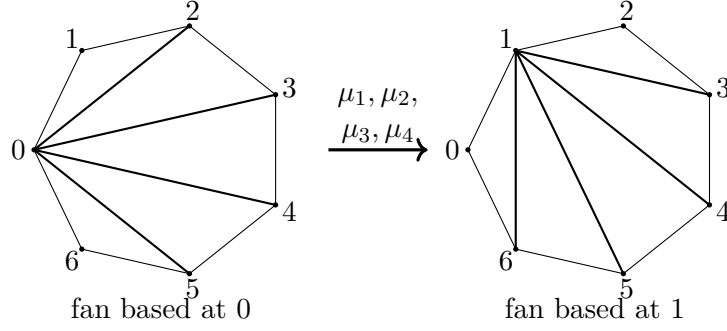


Figure 3: For type A_4 , the left-to-right source sweep flips $(0, 2), (0, 3), (0, 4), (0, 5)$ and replaces them by $(1, 3), (1, 4), (1, 5), (1, 6)$.

where an interval denotes both a denominator interval and a positive root.

Lemma 6.8 (A source sweep rotates a fan). *Let $N = n + 3$, and read polygon vertices modulo N . Suppose a triangulation is the fan based at a vertex q , with its mutable diagonals listed in the order*

$$d_k = (q, q + k + 1), \quad 1 \leq k \leq n.$$

Flipping $d_1, d_2, \dots, d_n$ in this order replaces d_k by

$$d'_k = (q + 1, q + k + 2)$$

at the k th step. After the full sequence, the triangulation is the fan based at $q + 1$.

Proof. Before the first flip, $d_1 = (q, q + 2)$ is the diagonal of the quadrilateral with vertices $q, q + 1, q + 2, q + 3$, so it is replaced by $(q + 1, q + 3)$. Assume that the first $k - 1$ flips have introduced

$$(q + 1, q + 3), \dots, (q + 1, q + k + 1).$$

Then $d_k = (q, q + k + 1)$ is adjacent to the two triangles with third vertices $q + 1$ and $q + k + 2$. Hence its flip is $(q + 1, q + k + 2)$. This proves the formula by induction on k . The resulting n diagonals are precisely all diagonals from $q + 1$ to its nonadjacent vertices, so they form the fan based at $q + 1$. $\square$

Proposition 6.9. *During the r th source sweep, for $1 \leq r \leq n$, the first $n - r + 1$ mutations introduce the diagonals*

$$(r, r + 2), (r, r + 3), \dots, (r, n + 2). \quad (23)$$

Their denominator intervals are respectively

$$[r, r], [r, r + 1], \dots, [r, n].$$

Consequently, Keller's knitting algorithm produces every noninitial cluster variable.

Proof. By Lemma 6.8, the first sweep changes the fan based at 0 into the fan based at 1, the second changes the fan based at 1 into the fan based at 2, and so on. Thus the r th sweep starts from the

fan based at $r - 1$. Applying the formula in Lemma 6.8 with $q = r - 1$, its k th flip introduces

$$(r, r + k + 1)$$

with vertex labels read modulo $n + 3$.

For $1 \leq k \leq n - r + 1$, no wraparound occurs and $r + k + 1 \leq n + 2$, giving exactly the diagonals in (23). When $k > n - r + 1$, the second endpoint wraps around the polygon; those later flips complete the fan rotation and may reproduce diagonals encountered in earlier sweeps. They are not needed for the enumeration of new variables.

By Lemma 6.2, $(r, r + k + 1)$ crosses exactly

$$\tau_r, \tau_{r+1}, \dots, \tau_{r+k-1},$$

so its denominator interval is $[r, r + k - 1]$. Every interval $[i, j] \subseteq [1, n]$ occurs exactly once by taking $r = i$ and $k = j - i + 1$. Therefore the source sweeps produce all $n(n + 1)/2$ noninitial variables. The initial variables are present before the process begins, so the knitting algorithm is exhaustive. $\square$

Lemma 6.7 and Proposition 6.9 prove statement (v) of Theorem 6.1, and the theorem is complete.

Table 3: The interval dictionary used in the proof.

Cluster algebra	Polygon	Interval/root data
noninitial variable $X_{a,b}$	diagonal (a, b)	$[a, b - 2]$ and $\alpha_a + \dots + \alpha_{b-2}$
reduced denominator support	crossed fan diagonals	simple-root support of the same interval
knitting/source sweep	rotation of a fan	enumeration of all intervals $[i, j]$

7 Conclusion

For the linearly oriented quiver Q_n , the type A_n polygon model turns Keller's Theorem 3.1 into explicit combinatorics. Quiver mutation is a local flip. Cluster variables are diagonals. Relative to the initial fan, a noninitial diagonal crosses a unique consecutive set of fan diagonals. The same interval indexes a positive root of $\mathfrak{sl}_{n+1}$. The Ptolemy recurrence produces a Laurent expression whose denominator is the product of the crossed fan variables, and the explicit numerator formula proves that none of those factors cancels. This yields the Laurent property, the denominator-root correspondence, and the total count $n(n + 3)/2$.

The knitting algorithm has an equally concrete interpretation: repeated left-to-right source sweeps rotate the fan one vertex at a time. The newly created diagonals enumerate all intervals $[i, j]$, and therefore all positive roots and all noninitial cluster variables. The proof is special to type A in its elementary form. Other Dynkin types still satisfy Keller's theorem, but they do not admit the same unpunctured-polygon model, and their complete treatment requires either different combinatorics or the deeper representation-theoretic machinery surveyed by Keller.

References

- [1] P. Etingof, *18.745: Lie Groups and Lie Algebras I*, MIT lecture notes, PDF version 2021.
- [2] S. Fomin and A. Zelevinsky, *Cluster algebras I: Foundations*, J. Amer. Math. Soc. **15** (2002), no. 2, 497–529.
- [3] S. Fomin and A. Zelevinsky, *Cluster algebras II: Finite type classification*, Invent. Math. **154** (2003), no. 1, 63–121.
- [4] B. Keller, *Cluster Algebras, Quiver Representations and Triangulated Categories*, in *Triangulated Categories*, London Math. Soc. Lecture Note Ser. **375**, Cambridge Univ. Press, 2010, pp. 76–160; arXiv:0807.1960 (supplied revision last modified December 8, 2020).
- [5] L. K. Williams, *Cluster Algebras: An Introduction*, Bull. Amer. Math. Soc. **51** (2014), no. 1, 1–26.